

Departement Elektriese, Elektroniese en
Rekenaar-Ingenieurswese

Department of Electrical, Electronic and
Computer Engineering

Eksamen

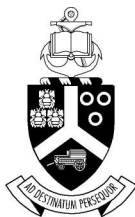
Kopiereg voorbehou

Examination

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Module EBN122
Elektrisiteit en Elektronika
13 November 2009

Module EBN122
Electricity and Electronics
13 November 2009



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Eksamensinligting:

Exam information:

Maksimum punte: <i>Maximum marks:</i>	90	Volpunte: <i>Full marks:</i>	90
Duur van vraestel: <i>Duration of paper:</i>	180 minute <i>180 minutes</i>	Oopboek / toeboek: <i>Open / closed book:</i>	Toe <i>Closed</i>
Totale aantal bladsye (hierdie blad ingesluit): <i>Total number of pages (including this page):</i>	30		

BELANGRIK- IMPORTANT

- Die eksamenregulasies van die Universiteit van Pretoria geld.
The examination regulations of the University of Pretoria apply.
- Alle vrae moet op die ANTWOORDBLAD beantwoord word.
All Questions must be answered on the ANSWER SHEET.
- Studente mag hul eie rowwe berekeninge op die vraestel doen – die vraestel word nie ingeneem nie.
Students may do their own rough calculations on the question paper – the question paper will not be taken in.
- Polêre vorm: $A \angle \theta$; Reghoekige vorm: $a + jb$
Polar form: $A \angle \theta$; Rectangular form: $a + jb$

Dosent(e): <i>Lecturer(s):</i>	1 J Joubert
	2 D Bhatt
	3 B Mabuza

Groephoof: Ek bevestig dat die vraestel die uitkomstes toets soos gespesifiseer in die studiehandleiding.

Group head: I confirm that the question paper evaluates the outcomes as specified in the study guide.

Groephoof: <i>Group Head:</i>	Prof J Joubert	Handtekening: <i>Signature:</i>	
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VRAAG/QUESTION 1 [22]

Vraag 1.1

Question 1.1

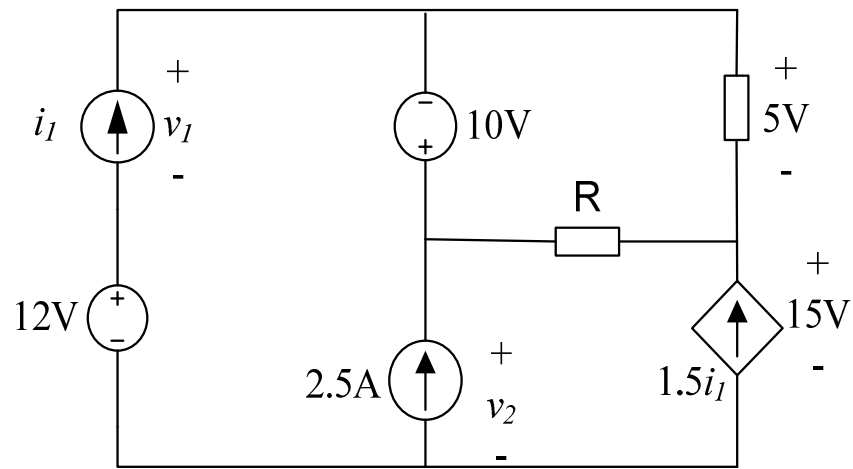


Fig. 1.1

Bepaal die spannings v_I , v_2 , en die stroom i_I . [3]

Determine the voltages v_I , v_2 and the current i_I . [3]

Vraag 1.2	Question 1.2
Bereken die totale lading deur 'n toestel by $t = 2\text{s}$, as die stroom deur die toestel gegee word deur die uitdrukking: $i(t) = 5 (1 - e^{-2t}) \text{ mA}$. [2]	Calculate the total charge through a device at $t = 2\text{s}$, when the current through the device is given by the expression: $i(t) = 5 (1 - e^{-2t}) \text{ mA}$. [2]

Vraag 1.3

Question 1.3

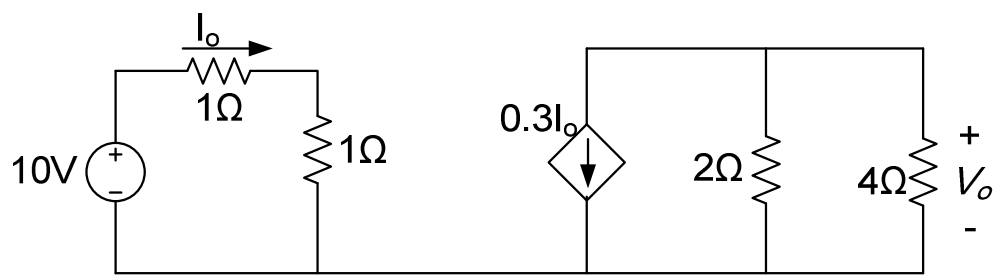


Fig. 1.3

Bereken I_o , en V_o ? [2]

Calculate I_o , and V_o ? [2]

Vraag 1.4

Question 1.4

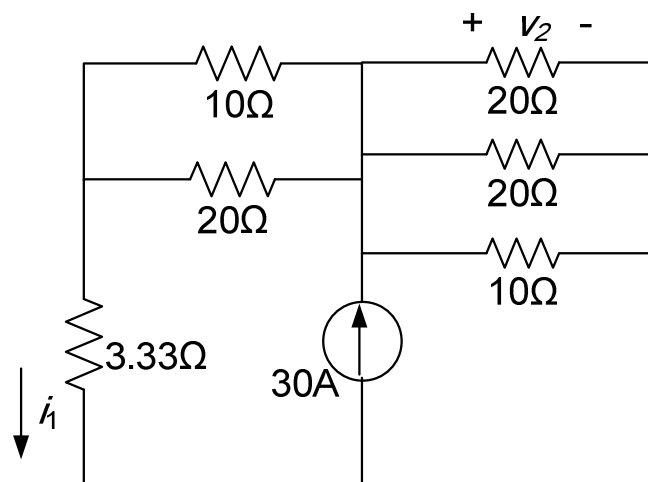


Fig. 1.4

Bereken die volgende: [3]

- (a) i_1
- (b) v_2 en
- (c) die drywing ge-assosieer met die 30A stroombron?

Calculate the following: [3]

- (a) i_1
- (b) v_2 and
- (c) the power associated with 30A current source?

Vraag 1.5

Question 1.5

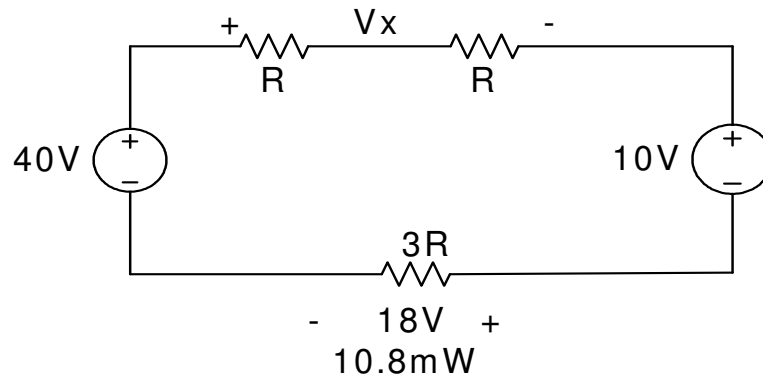


Fig. 1.5

Bereken:

(a) R [2]

(b) V_x [1]

Calculate:

(a) R [2]

(b) V_x [1]

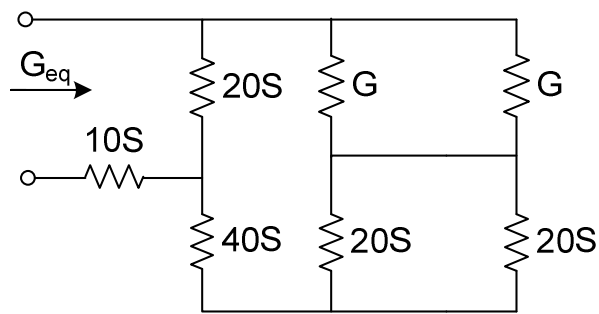
Vraag 1.6
Question 1.6


Fig. 1.6

Bepaal [2]

- (a) G_{eq} as $G = 0$
- (b) G_{eq} as $G = \infty$

Determine [2]

- (a) G_{eq} when $G = 0$
- (b) G_{eq} when $G = \infty$

Vraag 1.7

Question 1.7

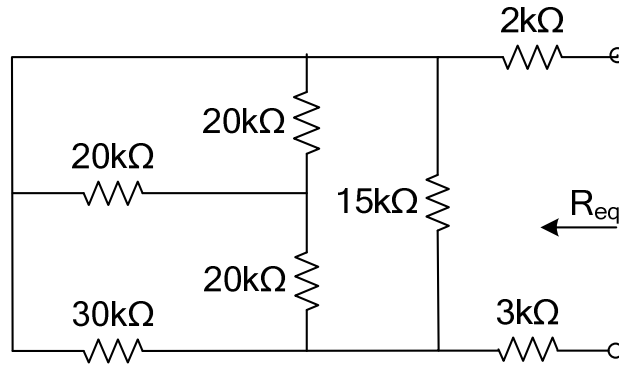


Fig. 1.7

Bereken R_{eq} [2]

Calculate R_{eq} [2]

Vraag 1.8

Question 1.8

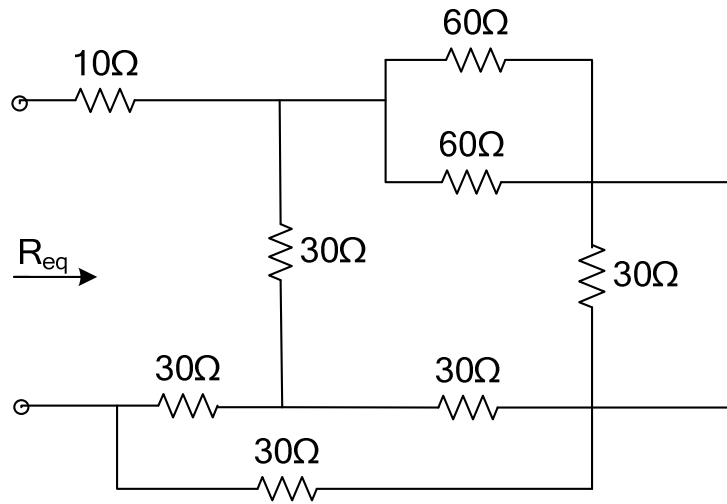


Fig. 1.8

Calculate R_{eq} [2]

Calculate R_{eq} [2]

Vraag 1.9

Question 1.9

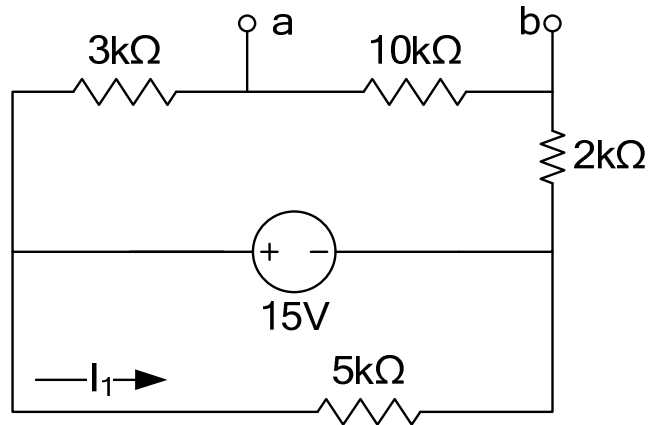


Fig. 1.9

(a) Bepaal V_{ab} en I_1 ? [2]

(b) As die 15V bron afgeskakel sou word, wat is die weerstand oor terminale $a-b$? [1]

(a) Determine V_{ab} and I_1 ? [2]

(b) If 15V source voltage is switched off what is the resistance across terminals $a-b$? [1]

VRAAG/QUESTION 2 [24]

Vraag 2.1

Question 2.1

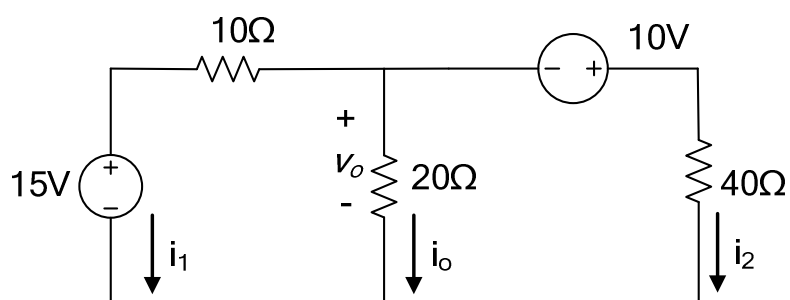


Fig. 2.1

Gebruik node-analise en bereken v_o [2]

Using nodal analysis calculate v_o [2]

Vraag 2.2	Question 2.2
Bepaal die determinant Δ en los die matriksvergelyking op om v_2 te bepaal. [2] $\begin{bmatrix} 3 & 2 & -1 \\ -1 & 3 & -2 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 10 \\ -5 \\ 0 \end{bmatrix}$	Find the determinant Δ and solve the matrix equation to find v_2. [2] $\begin{bmatrix} 3 & 2 & -1 \\ -1 & 3 & -2 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 10 \\ -5 \\ 0 \end{bmatrix}$

Vraag 2.3

Question 2.3

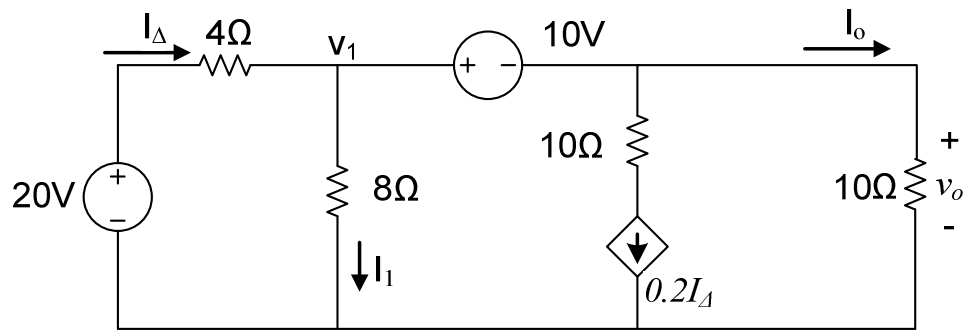


Fig. 2.3

Gebruik node-analise om die stroombaan op te los.
Bepaal v_o en v_I [3]

Use nodal analysis to solve the circuit and find the
values of v_o and v_I [3]

Vraag 2.4

Question 2.4

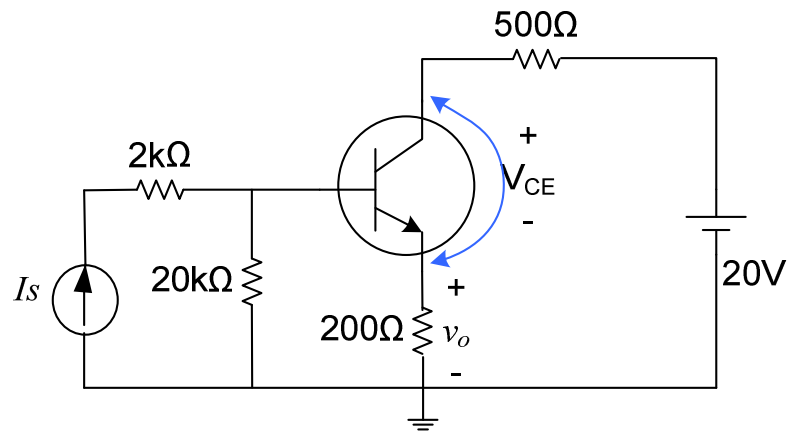


Fig. 2.4

$\beta = 110$ en $v_{BE} = 0.75\text{V}$ vir die bostaande stroombaan.

- (a) Bepaal I_s as $v_o = 5.5\text{V}$, en skryf jou antwoord in μA . [2]
 (b) Bepaal v_{CE} as $v_o = 4.5\text{V}$. [2]

$\beta = 110$ and $v_{BE} = 0.75\text{V}$ for the circuit above.

- (a) Determine I_s if $v_o = 5.5\text{V}$, and express your answer in μA . [2]
 (b) Determine v_{CE} if $v_o = 4.5\text{V}$. [2]

Vraag 2.5

Question 2.5

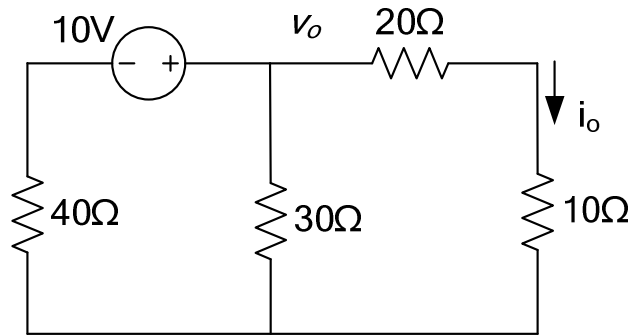


Fig. 2.5

- (a) Bereken stroom i_o in die stroombaan [2]
 (b) Wat sal die waarde van i_o wees as die spanningsbron na 20V verander word [1]

- (a) Calculate current i_o in the circuit [2]
 (b) What will the value of this current i_o be when the source voltage is raised to 20V [1]

Vraag 2.6

Question 2.6

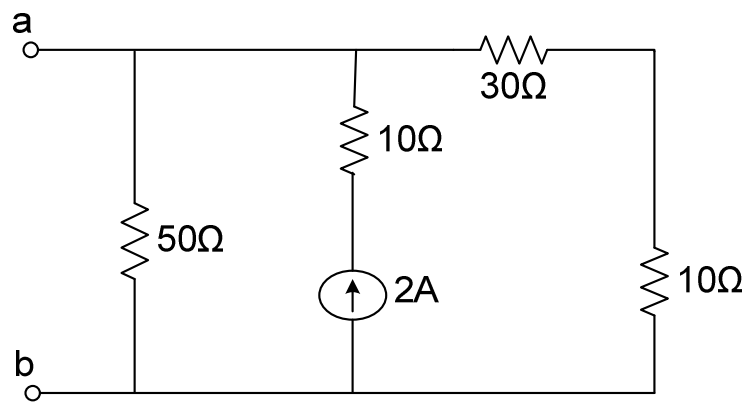


Fig. 2.6

Bepaal die waarde van 'n weerstand wat jy oor terminale a - b kan plaas sodat maksimum drywing na die weerstand oorgedra word. [1]

Determine the value of a resistor placed across the open terminals, a - b , resulting in maximum power transferred to the resistor. [1]

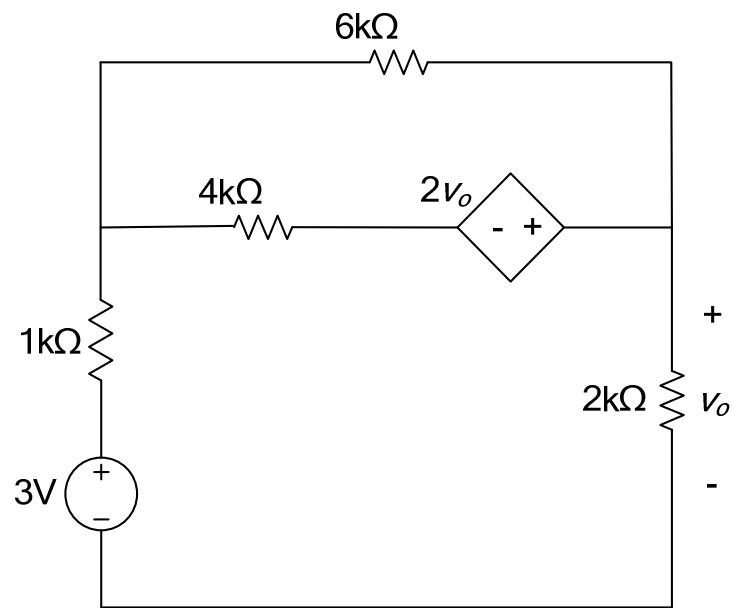


Fig. 2.7

Gebruik brontransformaties om v_o te bepaal [3]Use source transformation to find v_o [3]

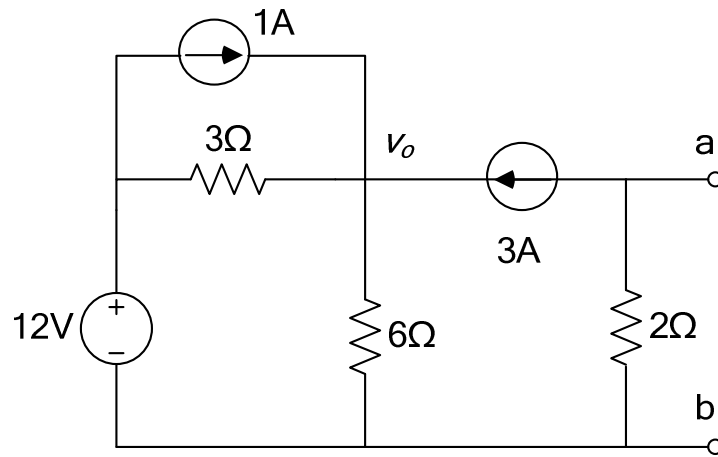


Fig. 2.8

Bepaal R_{th} and V_{TH} met verwysing tot terminale a–b. [2]Find R_{th} and V_{TH} with respect to terminals a–b. [2]

Vraag 2.9

Question 2.9

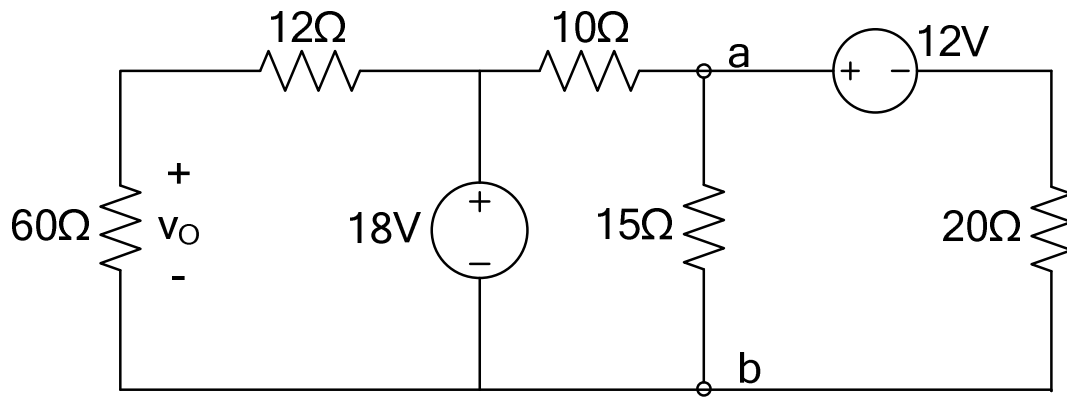


Fig. 2.9

Bepaal V_{TH} met verwysing tot terminale a–b. [2]
(verwyder die 15Ω las)

Find V_{TH} with respect to terminals a–b. [2]
(remove the 15Ω load)

Vraag 2.10

Question 2.10

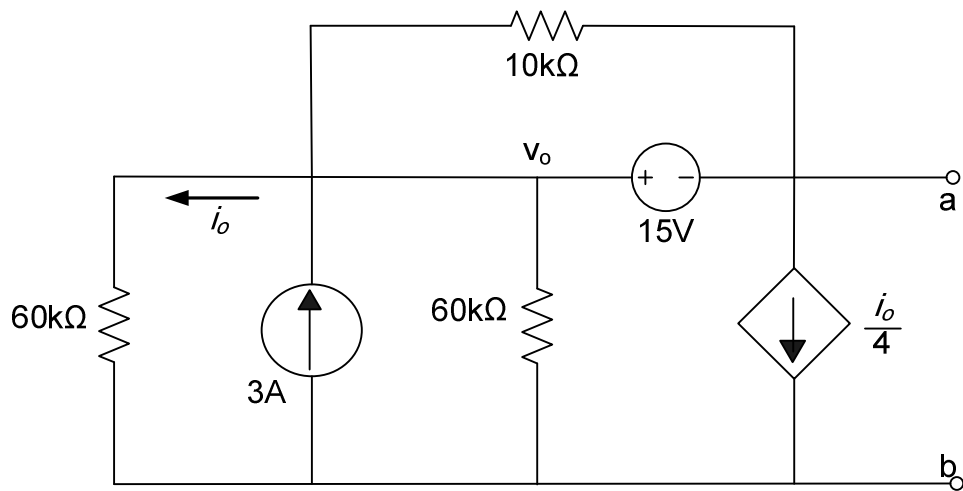
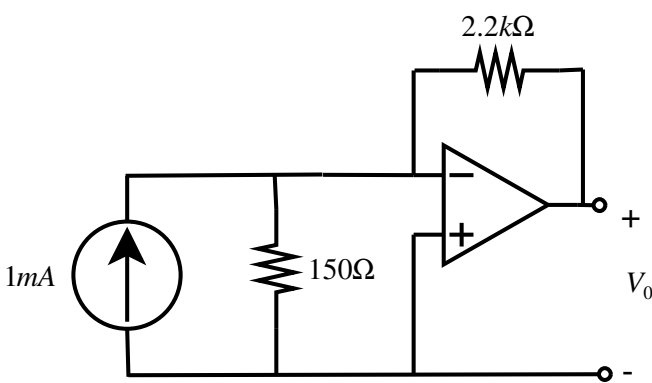
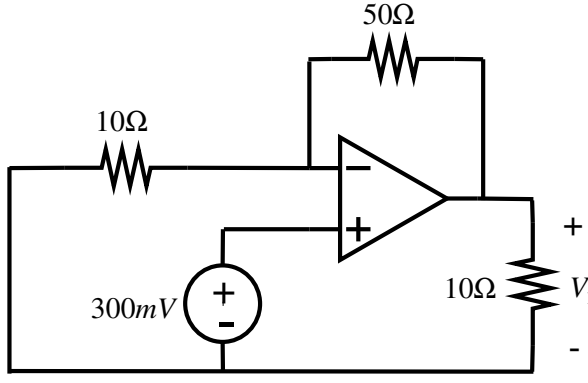


Fig. 2.10

Bepaal die Thevenin-ekwivalente weerstand R_{TH} met verwysing tot terminale a-b. [2]

Determine the Thevenin equivalent resistance R_{TH} with respect to terminals a-b. [2]

VRAAG 3 – QUESTION 3 [20]

Vraag 3.1	Question 3.1
 Fig. 3.1(a)	 Fig. 3.1(b)
(a) Bepaal V_o in Fig. 3.1(a) [2] (b) Bepaal V_i in Fig. 3.1(b) [2]	(a) Determine V_o in Fig. 3.1(a) [2] (b) Determine V_i in Fig. 3.1(b) [2]

Vraag 3.2

Question 3.2

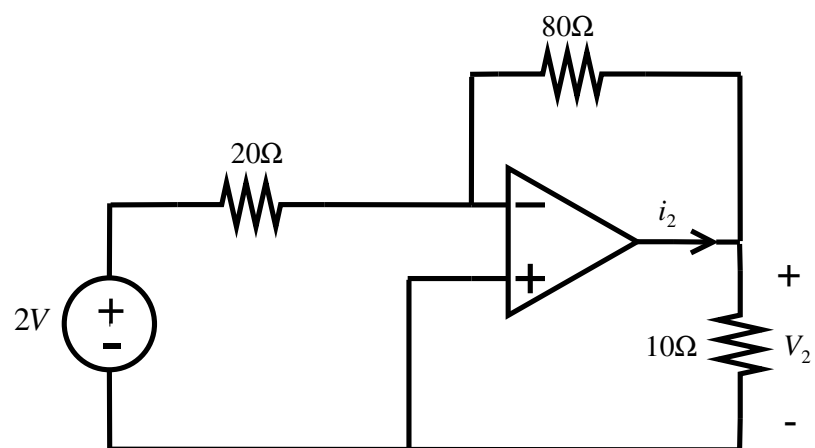


Fig. 3.2

(a) Bepaal V_2 in Fig. 3.2 [2]

(b) Bepaal i_2 in Fig. 3.2 [2]

(a) Determine V_2 in Fig. 3.2 [2]

(b) Determine i_2 in Fig. 3.2 [2]

Vraag 3.3

Question 3.3

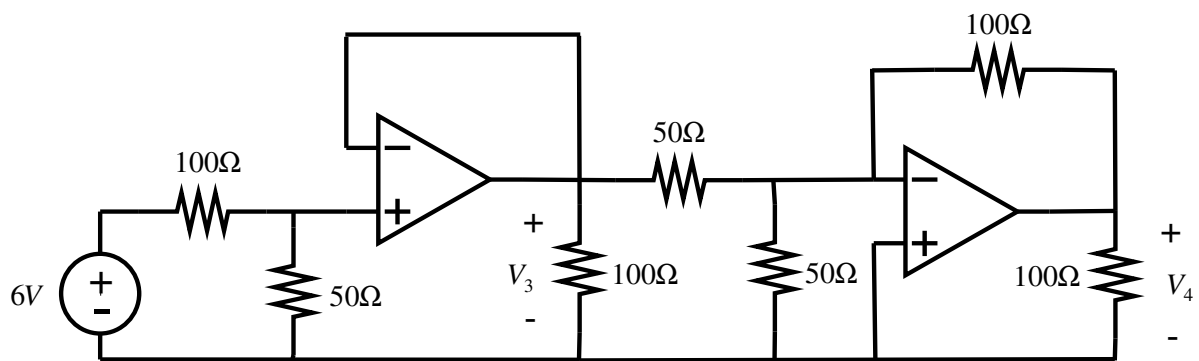


Fig. 3.3

(a) Bepaal V_3 in Fig. 3.3 [2]

(b) Bepaal V_4 in Fig. 3.3 [2]

(a) Determine V_3 in Fig. 3.3 [2]

(b) Determine V_4 in Fig. 3.3 [2]

Vraag 3.4	Question 3.4
(a) Die stroom deur 'n 1 mF kapasitor is $7\sin(2t)$ mA vir $t \geq 0$ s. Bepaal die spanning oor die kapasitor by $t = 0.1$ s. Aanvaar $v(0) = 0$ V. [2] (b) Die stroom deur 'n 10 mH induktor is $4e^{-6t}$ mA vir $t \geq 0$ s. Bepaal die spanning $v(t)$ oor die induktor. [2]	(a) The current through a 1 mF capacitor is $7\sin(2t)$ mA for $t \geq 0$ s. Determine the voltage across the capacitor at $t = 0.1$ s. Assume $v(0) = 0$ V. [2] (b) The current through a 10 mH inductor is $4e^{-6t}$ mA for $t \geq 0$ s. Determine the voltage $v(t)$ across the inductor. [2]

Vraag 3.5

Question 3.5

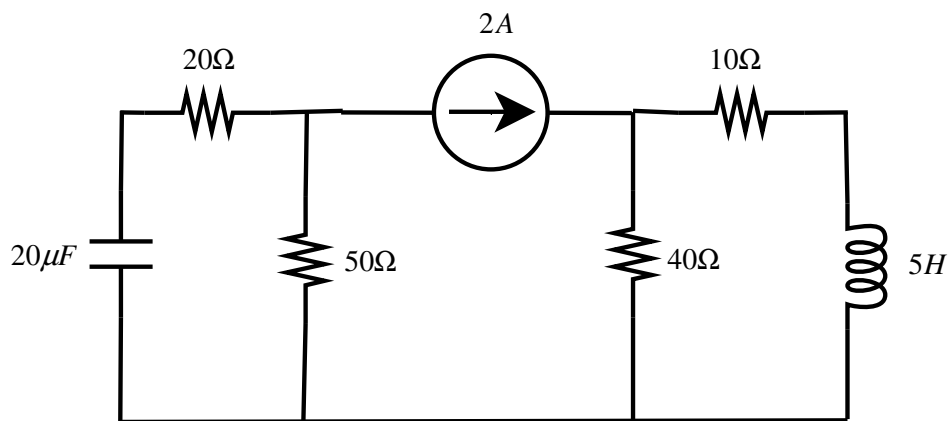


Fig. 3.5

Bepaal W_L en W_C , die energie gestoor in die induktor en kapasitor onderskeidelik. [4]

Determine W_L and W_C , the energy stored in the inductor and capacitor respectively. [4]

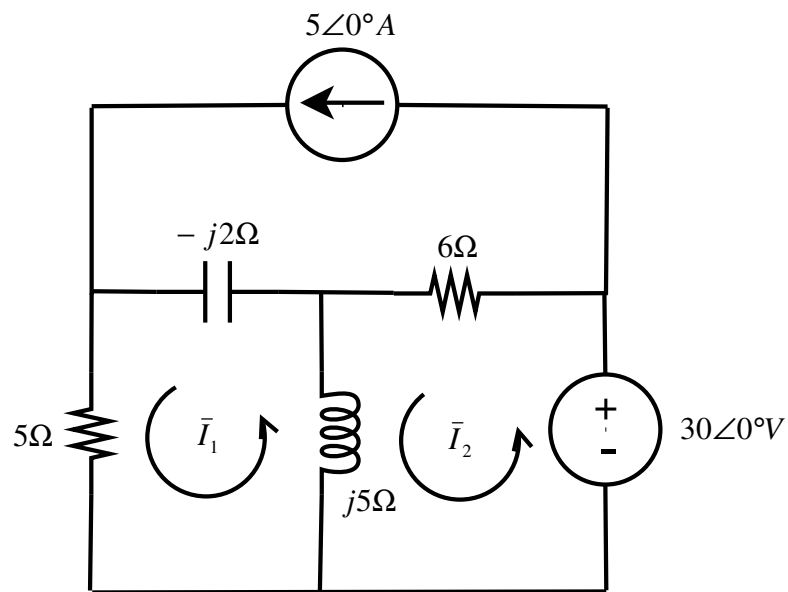
VRAAG 4 – QUESTION 4 [24]

Vraag 4.1	Question 4.1
(a) Skryf as 'n enkele cosinus uitdrukking: $v(t) = 14.14\cos(\omega t + 45^\circ) + 10\cos(\omega t - 90^\circ)$ V [2] (b) Watter van die volgende fasors loop voor? $\bar{V}_1 = 10\angle 30^\circ$, $\bar{V}_2 = 15\angle -210^\circ$ [2] (c) Bepaal $\bar{V}_1$ (in reghoekige vorm): $\begin{bmatrix} 1+j1 & -2 \\ -1 & 2-j2 \end{bmatrix} \begin{bmatrix} \bar{V}_1 \\ \bar{V}_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ [2]	(a) Write as a single cosine expression: $v(t) = 14.14\cos(\omega t + 45^\circ) + 10\cos(\omega t - 90^\circ)$ V [2] (b) Which of the following phasors is leading? $\bar{V}_1 = 10\angle 30^\circ$, $\bar{V}_2 = 15\angle -210^\circ$ [2] (c) Determine $\bar{V}_1$ (in rectangular form): $\begin{bmatrix} 1+j1 & -2 \\ -1 & 2-j2 \end{bmatrix} \begin{bmatrix} \bar{V}_1 \\ \bar{V}_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ [2]

Vraag 4.2	Question 4.2
(a) Evalueer die volgende (skryf antwoord in reghoekige vorm): $\frac{(10 + j20)^2}{(3 + j4)} \quad [2]$ (b) Evalueer die volgende (skryf antwoord in polêre vorm): $\frac{(14 - j9)}{(-2 + j8)} + 2e^{j30^\circ} \quad [2]$	(a) Evaluate the following (write the answer in rectangular form): $\frac{(10 + j20)^2}{(3 + j4)} \quad [2]$ (b) Evaluate the following (write the answer in polar form): $\frac{(14 - j9)}{(-2 + j8)} + 2e^{j30^\circ} \quad [2]$

Vraag 4.3

Question 4.3



Gebruik lus-analise om twee gelyktydige vergelykings vir fasors $\bar{I}_1$ en $\bar{I}_2$ op te stel. Skryf in die vorm

$$\bar{a}\bar{I}_1 + \bar{b}\bar{I}_2 = \bar{c}$$

$$\bar{d}\bar{I}_1 + \bar{e}\bar{I}_2 = \bar{f}$$

met $\bar{a}$, $\bar{b}$, $\bar{c}$, $\bar{d}$, $\bar{e}$ en $\bar{f}$ komplekse koeffisiënte in reghoekige vorm. [4]

Use mesh analysis to set up two simultaneous equations for phasors $\bar{I}_1$ and $\bar{I}_2$. Write in the form

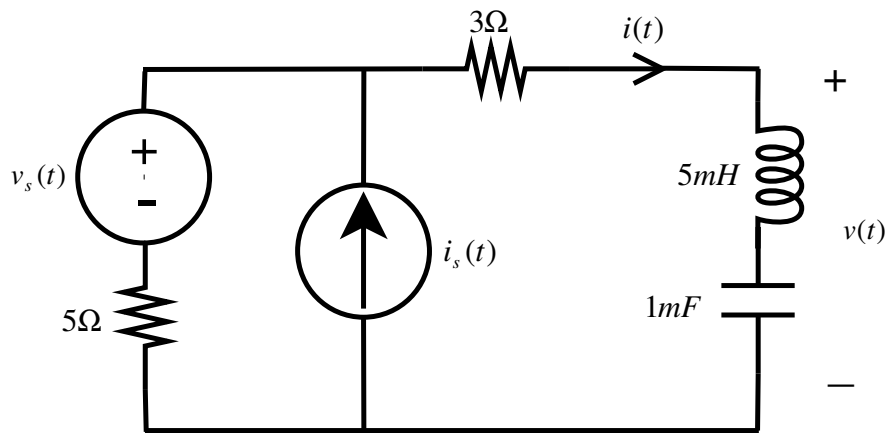
$$\bar{a}\bar{I}_1 + \bar{b}\bar{I}_2 = \bar{c}$$

$$\bar{d}\bar{I}_1 + \bar{e}\bar{I}_2 = \bar{f}$$

with $\bar{a}$, $\bar{b}$, $\bar{c}$, $\bar{d}$, $\bar{e}$ and $\bar{f}$ complex coefficients in rectangular form. [4]

Vraag 4.4

Question 4.4



Gebruik die beginsel van superposisie:

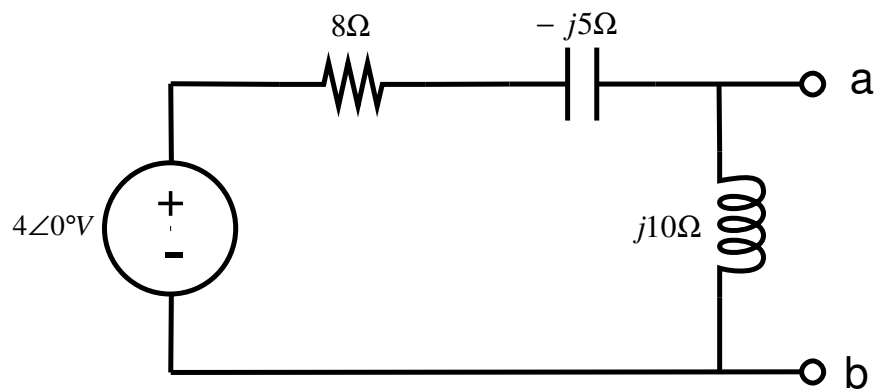
- (a) Bereken $i(t)$ slegs agv die spanningsbron
 $v_s(t) = 10\cos(200t + 30^\circ)$ [2]
- (b) Bereken $v(t)$ slegs agv die spanningsbron
 $v_s(t) = 10\cos(200t + 30^\circ)$ [2]
- (c) Bereken $i(t)$ slegs agv die stroombron
 $i_s(t) = 5\cos(200t)$ [2]

Use the principle of superposition:

- (a) Determine $i(t)$ only due to the voltage source
 $v_s(t) = 10\cos(200t + 30^\circ)$ [2]
- (b) Determine $v(t)$ only due to the voltage source
 $v_s(t) = 10\cos(200t + 30^\circ)$ [2]
- (c) Determine $i(t)$ only due to the current source
 $i_s(t) = 5\cos(200t)$ [2]

Vraag 4.5

Question 4.5



Bepaal die Norton ekwivalent tov terminale a-b. Skryf die antwoorde in polêre vorm. [4]

Find the Norton equivalent wrt terminals a-b. Write the answers in polar form. [4]

EINDE - END

EBN122 EKSAMEN ANTWOORDBLAD/ EXAM ANSWER SHEET

13 NOVEMBER 2009

Student se besonderhede/ <i>Student's details</i>									
Van en voorletters: <i>Surname and initials:</i>					Studentenommer: <i>Student number:</i>				
Graadkursus: <i>Degree Course:</i>									
Tel. nr.: <i>Tel no.:</i>									

PUNTE/MARKS									
Q1	/22	Q2	/24	Q3	/20	Q4	/24	Final	/90

VRAAG /QUESTION 1 [22]			
1.1	$v_I =$	$v_2 =$	$i_I =$
1.2	$Q =$		
1.3	$I_o =$	$V_o =$	
1.4	$i_I =$	$v_2 =$	$P =$
1.5	$R =$	$V_x =$	
1.6	$G_{eq} =$	$G_{eq} =$	
1.7	$R_{eq} =$		
1.8	$R_{eq} =$		
1.9	$V_{ab} =$	$I_I =$	$R_{ab} =$

VRAAG /QUESTION 2 [24]	
2.1	$v_o =$
2.2	$\Delta =$ $v_2 =$
2.3	$v_o =$ $v_I =$
2.4	$I_s =$ $V_{CE} =$
2.5	$i_o =$ $i_o =$
2.6	$R_{ab} =$
2.7	$v_o =$
2.8	$R_{TH} =$ $V_{TH} =$
2.9	$V_{TH} =$
2.10	$R_{TH} =$

VRAAG 3 – QUESTION 3 [20]		
3.1	(a) $V_0 =$	(b) $V_I =$
3.2	(a) $V_2 =$	(b) $i_2 =$
3.3	(a) $V_3 =$	(b) $V_4 =$
3.4	(a) $v(0.1s) =$	(b) $v(t) =$
3.5	$W_L =$	$W_C =$

VRAAG 4 – QUESTION 4 [24]			
4.1	(a) $v(t) =$		
	(b)		
	(c) $\bar{V}_1 =$		
4.2	(a)		
	(b)		
4.3	$\bar{a} =$	$\bar{b} =$	$\bar{c} =$
	$\bar{d} =$	$\bar{e} =$	$\bar{f} =$
4.4	(a) $i(t) =$		
	(b) $v(t) =$		
	(c) $i(t) =$		
4.5	$\bar{I}_N =$	$\bar{Z}_N =$	